

Joji Morikawa

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Education

University of California Los Angeles, BS in Computer Engineering Sept 2025 - Expected Fall 2028

- GPA: 3.8

Experience

Embodied AI intern, GODOT - Kobe, Japan June 2026 - Aug 2026

- Developed C++ firmware for an autonomous mobile robot.
- Created middleware to interface between the ESP32 and NVIDIA Jetson, defining ROS 2 topics and data flows connecting low-level control and sensor feedback with localization, LiDAR, and autonomous navigation.
- Designed closed-loop wheel velocity control, and motion-adaptive odometry covariance to account for skid-steer wheel slip during localization.
- Built a debugging and telemetry pipeline using custom ROS 2 topics.
- Built a ROS 2/Gazebo Software-in-the-Loop (SIL) environment to validate motion control, odometry, localization, and sensor integration before hardware deployment.

Projects

Rocket Avionics, Live Telemetry  RF Communication, PCB Design

- Designed a custom 2-layer PCB for model rocket nose cone, and ground station, integrating SAMD21 microcontroller, RFM95W LoRa radio transceiver, and telemetry sensors.
- Developed embedded C++ firmware for both transmitter and ground-station receiver, implementing radio communication, telemetry acquisition, packet transmission, and data logging.
- Implemented low-power mode, pre-launch operation with MCU sleep/wake behavior and radio-based activation. Streamed live altitude telemetry and logged flight data to CSV for post-flight analysis.
- Reached 3,300-ft apogee, with uninterrupted telemetry until impact.

FPGA Real-Time Game System Xilinx Vivado, Verilog

- Built an interactive game on a Xilinx Basys 3 FPGA, implementing hardware logic in Verilog
- Designed modular synchronous logic including state machines, clock dividers, debouncers, edge detection, counters, collision logic, and combinational display decoders.
- Verified and debugged the design through HDL simulation and hardware testing

STM 32 motion recognition Embedded Systems, ML

- Adapted an existing STM32-based gesture recognition firmware to classify alphabetic character motions.
- Created an interactive embedded spelling-game application to test and demonstrate real-time gesture classification.

Technologies

Languages: C++, C, Python, Verilog, Java, JavaScript, x86 Assembly (MASM), SQL

Technologies: Linux, ROS2, micro-ROS, Gazebo, PlatformIO, Xilinx Vivado, Git, SolidWorks, MATLAB, Fusion360, KiCad

Interests

UCLA Track and Field club, IEEE, community college outreach